Spinning top research --- Solution

X25 (2025) — English translation

A1 **Write down in vector form the expression for the instantaneous speed $\vec{v}$ of fragment dm at the moment under consideration. Express your answer using $\vec{\Omega}$, $\vec{r}$.** **0.20**

We use Euler's formula for a rigid body.

Answer:

$$\vec{v} = [\vec{\Omega} \times \vec{r}]$$

A2 **Write down in vector form the expression for the angular momentum $d\vec{L}$ of the fragment dm at the moment under consideration. Express your answer using $\vec{\Omega}$, $\vec{r}$, dm.** **0.20**

By definition, the angular momentum of an element of mass dm:

Answer:

$$d\vec{L} = dm[\vec{r} \times [\vec{\Omega} \times \vec{r}]]$$

A3 **Obtain expressions for the components of the vector $d\vec{L} = (dL_x, dL_y, dL_z)$ in the coordinate system $OXYZ$. Express your answer using X, Y, Z, Ω_x, Ω_y, Ω_z, dm.** **0.30**

Let's expand the double vector product in the expression for $d\vec{L}$:

$$d\vec{L} = dm(\vec{r} \times (\vec{\Omega} \times \vec{r})) = dm(\vec{\Omega} r^2 - \vec{r}(\vec{\Omega} \cdot \vec{r})).$$

Распишем полученное векторное выражение покомпонентно:

$$dL_x = dm\left(\Omega_x(X^2+Y^2+Z^2) - X(\Omega_x X + \Omega_y Y + \Omega_z Z)\right), dL_y = dm\left(\Omega_y(X^2+Y^2+Z^2) - Y(\Omega_x X + \Omega_y Y + \Omega_z Z)\right), dL_z = dm\left(\Omega_z(X^2+Y^2+\right.$$

Simplifying, we get the answer.

Answer:

$$dL_x = dm\left(\Omega_x(Y^2+Z^2) - \Omega_y XY - \Omega_z XZ\right) dL_y = dm\left(-\Omega_x YX + \Omega_y(X^2+Z^2) - \Omega_z YZ\right) dL_z = dm\left(-\Omega_x ZX - \Omega_y ZY + \Omega_z(X^2+Y^2)\right)$$

A4 **For arbitrary motion of a rigid body, we can write angular momentum in the form** **0.50**

$$L_x = I_{xx}\Omega_x + I_{xy}\Omega_y + I_{xz}\Omega_z;$$
$$L_y = I_{yx}\Omega_x + I_{yy}\Omega_y + I_{yz}\Omega_z;$$
$$L_z = I_{zx}\Omega_x + I_{zy}\Omega_y + I_{zz}\Omega_z.$$

Obtain expressions for the coefficients I_{xx}, I_{xy}, I_{xz}, I_{yx}, I_{yy}, I_{yz}, I_{zx}, I_{zy}, I_{zz} in the form of integrals over dm. The integral over dm is the sum over infinitesimal regions of a solid body of mass dm. For example, the coordinates of the center of mass can be written as

$$X_c = \frac{1}{m}\int X dm; \quad Y_c = \frac{1}{m}\int Y dm, \quad Z_c = \frac{1}{m}\int Z dm.$$

In further paragraphs, the formulas obtained here are not used.

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In further paragraphs the formulas obtained here are not used.

The answer is obtained by integrating over dm the expressions of the previous paragraph.

Answer:

$$I_{xx} = \int dm(Y^2+Z^2) \quad I_{yy} = \int dm(X^2+Z^2) \quad I_{zz} = \int dm(X^2+Y^2) I_{xy} = I_{yx} = -\int dm XY \quad I_{yz} = I_{zy} = -\int dm YZ \quad I_{xz} = I_{zx}$$

A5 **Derive expressions for $\vec{L}$ and T in the principal axes. Express your answer using I_x, I_y, I_z, ω_x, ω_y, ω_z, $\vec{e}_x$, $\vec{e}_y$, $\vec{e}_z$.** **0.50**

In the principal axes, the expressions for the angular momentum components $\vec{L}$ take the form

$$L_x = I_x\omega_x, \qquad L_y = I_y\omega_y, \qquad L_z = I_z\omega_z.$$

This gives the answer.

According to the condition

$$T = \frac{1}{2}\vec{\omega}\cdot\vec{L}.$$

Используя найденное выше выражение для $\vec{L}$, we arrive at the answer.

Answer:

$$\vec{L} = I_x\omega_x\vec{e}_x + I_y\omega_y\vec{e}_y + I_z\omega_z\vec{e}_z$$

A6 **Obtain expressions for the components of the vector $\dot{\vec{L}}$ in the moving coordinate system $\left(\dot{\vec{L}}\right)_x$, $\left(\dot{\vec{L}}\right)_y$, $\left(\dot{\vec{L}}\right)_z$. Express your answer using I_x, I_y, I_z, ω_x, ω_y, ω_z, $\dot{\omega}_x$, $\dot{\omega}_y$, $\dot{\omega}_z$.** **0.50**

Let us differentiate the previously obtained expression for $\vec{L}$. Since the unit vectors of the coordinate system are movable, they also need to be differentiated.

$$\dot{\vec{L}} = I_x\frac{d}{dt}(\omega_x\vec{e}_x) + I_y\frac{d}{dt}(\omega_y\vec{e}_y) + I_z\frac{d}{dt}(\omega_z\vec{e}_z) == I_x\dot{\omega}_x\vec{e}_x + I_y\dot{\omega}_y\vec{e}_y + I_z\dot{\omega}_z\vec{e}_z + I_x\omega_x\dot{\vec{e}}_x + I_y\omega_y\dot{\vec{e}}_y + I_z\omega_z\dot{\vec{e}}_z == I_x\dot{\omega}_x\vec{e}_x + I_y\dot{\omega}_y\vec{e}_y + I_z\dot{\omega}_z\vec{e}_z + (\vec{\omega}\times(I$$

Распишем в координатах векторное произведение:

$$[\vec{\omega}\times\vec{L}] = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \omega_x & \omega_y & \omega_z \\ L_x & L_y & L_z \end{vmatrix} = \vec{e}_x(\omega_y L_z - \omega_z L_y) + \vec{e}_y(\omega_z L_x - \omega_x L_z) + \vec{e}_z(\omega_x L_y - \omega_y L_z)$$

Таким образом получаем проекции $\dot{\vec{L}}$ на оси системы $Oxyz$.

Answer:

$$\left(\dot{\vec{L}}\right)_x = I_x\dot{\omega}_x + (I_z - I_y)\omega_y\omega_z \left(\dot{\vec{L}}\right)_y = I_y\dot{\omega}_y + (I_x - I_z)\omega_z\omega_x \left(\dot{\vec{L}}\right)_z = I_z\dot{\omega}_z + (I_y - I_x)\omega_x\omega_y$$

B1 **Find the projections of angular velocity ω_x, ω_y, ω_z. Express your answer using ω_s, $\dot{\theta}$, $\dot{\varphi}$, θ.** **0.40**

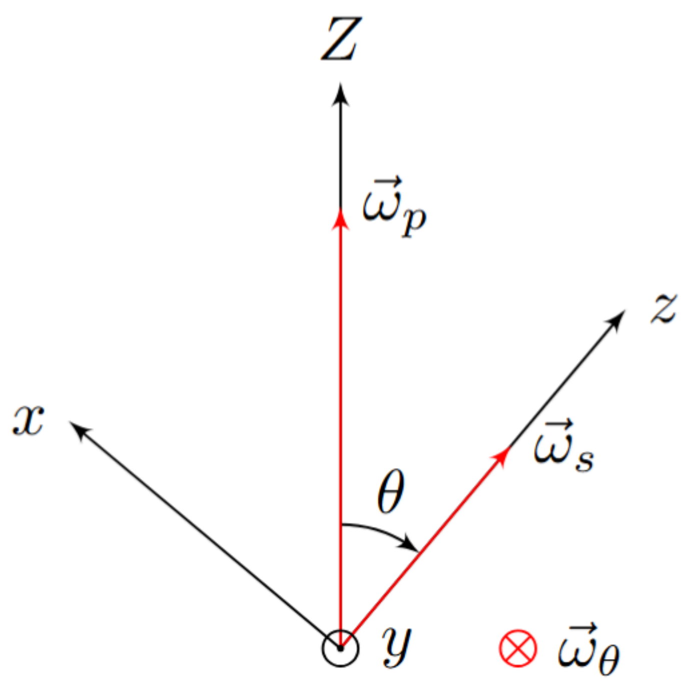
Projecting $\vec{\omega}_s, \vec{\omega}_\theta, \vec{\omega}_\varphi$ onto the axes x, y, z and taking into account

$$\vec{\omega}_\theta = -\dot{\theta}\vec{e}_y, \qquad \vec{\omega}_\varphi = \dot{\varphi}\vec{e}_Z,$$

we get the answer.

Answer:

$$\omega_x = \dot{\varphi}\sin\theta\omega_y = -\dot{\theta}\omega_z = \omega_s + \dot{\varphi}\cos\theta$$



B2 **Show that the projection L_1 of the angular momentum $\vec{L}$ onto the fixed vertical axis OZ is constant. Show that the projection L_2 of the angular momentum $\vec{L}$ onto the moving axis Oz is constant.** **0.40**

The moment of gravity is expressed by the formula

$$\vec{M} = \vec{l} \times m\vec{g},$$

из которой следует, что $\vec{M}$ перпендикулярен плоскости OZz. Из уравнения моментов в неподвижных осях следует

$$\dot{L}_1 = \dot{L}_Z = M_Z = 0.$$

Из уравнения Эйлера для оси Oz следует

$$I_z\dot{\omega}_z + (I_y - I_x)\omega_x\omega_y = 0.$$

В силу симметрии $I_x = I_y$, поэтому

$$\dot{L}_2 = I_z\dot{\omega}_z = 0.$$

Отметим, что перпендикулярности $\vec{M}$ и Oz для сохранения L_2 is not enough.

B3 **Using the results of step A5, express the values of the projections L_1 and L_2 in terms of ω_s, $\dot{\theta}$, $\dot{\varphi}$, θ, A, C.** **0.60**

Let us express L_1 through the projections $\vec{L}$:

$$L_1 = L_z\cos\theta + L_x\sin\theta = A\omega_x\sin\theta + C\omega_z\cos\theta.$$

Используя результат пункта B1, получаем окончательный ответ для L_1:

$$L_1 = C(\omega_s + \dot{\varphi}\cos\theta)\cos\theta + A\dot{\varphi}\sin^2\theta.$$

Для L_2 верно

$$L_2 = L_z = C\omega_z = C(\omega_s + \dot{\varphi}\cos\theta).$$

Answer:

$$L_1 = C(\omega_s + \dot{\varphi}\cos\theta)\cos\theta + A\dot{\varphi}\sin^2\theta \; L_2 = C(\omega_s + \dot{\varphi}\cos\theta)$$

B4 **At this point we will assume that $l = 0$, and the moment of gravity can be ignored. Let the angular momentum of the top be equal to L_0 and directed along the axis OZ. At time $t = 0$ the values of angles $\theta = \theta_0$, $\varphi = \varphi_0$. Find the dependence of angles θ, φ on time. This movement is called regular precession.** **0.70**

In the described case

$$L_1 = L_0, \qquad L_2 = L_0 \cos\theta.$$

В силу постоянства величин L_0, L_1, L_2 угол θ также остаётся постоянным. Используя выражения для L_1, L_2 из предыдущего пункта, получаем систему уравнений:

$$BEEQUATION$$

Решая систему, получаем

$$\dot\varphi = \frac{L_0}{A}$$

Далее зависимость φ from time to time turns out to be simple integration.

Answer:

$$\theta(t) = \theta_0 \varphi(t) = \varphi_0 + \frac{L_0}{A} t$$

C1 For an arbitrary position of the top, express the total energy E of the motion of the top (the sum of kinetic and potential) in terms of ω_s, $\dot\theta$, $\dot\varphi$, A, C, m, g, l, θ. **0.60**

Let's write down the expressions for potential and kinetic energies:

$$T = \frac{1}{2}(C\omega_z^2 + A(\omega_x^2 + \omega_y^2)) = \frac{1}{2}(C(\omega_s + \dot\varphi\cos\theta)^2 + A(\dot\theta^2 + \dot\varphi^2\sin^2\theta)), \Pi = mgl\cos\theta.$$

Полная энергия:

$$E = T + \Pi.$$

Answer:

$$E = \frac{1}{2}(C(\omega_s + \dot\varphi\cos\theta)^2 + A(\dot\theta^2 + \dot\varphi^2\sin^2\theta)) + mgl\cos\theta$$

C2 Express ω_s and $\dot\varphi$ in terms of L_1, L_2, A, C, θ using the results from step B3. **0.30**

The answer is obtained by solving the system for ω_s, $\dot\varphi$.

$$\begin{cases} L_1 = \dot\varphi(A\sin^2\theta + C\cos^2\theta) + C\omega_s\cos\theta \\ L_2 = C(\omega_s + \dot\varphi\cos\theta) \end{cases}$$

Answer:

$$\dot\varphi = \frac{L_1 - L_2\cos\theta}{A\sin^2\theta}\ \omega_s = \frac{L_2}{C} - \frac{L_1 - L_2\cos\theta}{A\sin^2\theta}\cos\theta$$

C3 Using the constancy L_1, L_2, reduce the expression for the total energy E to the form **0.60**

$$E = \frac{\mu}{2}\dot\theta^2 + U(\theta).$$

Получите выражения для μ, $U(\theta)$ через A, C, L_1, L_2, m,g, l. Note. As is known, the total energy is determined up to a constant, which for convenience can be set equal to zero.

Note. As is known, the total energy is determined up to a constant, which for convenience can be set equal to zero.

Substitution into the expression for the energy ω_s and $\dot\varphi$ from the previous paragraph leads to the result

$$E = \frac{A}{2}\dot\theta^2 + \frac{(L_1 - L_2\cos\theta)^2}{2A\sin^2\theta} + mgl\cos\theta + \frac{L_2^2}{2C}.$$

Последнее константное слагаемое можно отбросить:

$$E = \frac{A}{2}\dot\theta^2 + \frac{(L_1 - L_2\cos\theta)^2}{2A\sin^2\theta} + mgl\cos\theta.$$

The resulting expression yields the answer.

Answer:

$$\mu = AU(\theta) = mgl\cos\theta + \frac{(L_1 - L_2\cos\theta)^2}{2A\sin^2\theta}$$

D1 **Find the period of oscillation of the top. You will receive the answer in the form of a definite integral containing A, C, l, θ_0, g. Also obtain a simplified expression for the case $\pi - \theta_0 \ll 1$.** **0.80**

For given initial conditions

$$L_1 = L_2 = 0.$$

Закон сохранения энергии в этом случае запишется в виде

$$\frac{A\dot{\theta}^2}{2} + mgl\cos\theta = mgl\cos\theta_0.$$

Отсюда выражается зависимость $\dot{\theta}(\theta)$:

$$\dot{\theta} = \pm\sqrt{\frac{2mgl}{A}(\cos\theta_0 - \cos\theta)}.$$

Движение от $\theta = \theta_0$ до $\theta = \pi$ с $\dot{\theta} > 0$ занимает четверть периода. Для нахождения этого времени требуется проинтегрировать уравнение

$$dt = \frac{d\theta}{\dot{\theta}}.$$

Интегрируем:

$$\frac{T}{4} = \sqrt{\frac{A}{2mgl}}\int\limits_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0 - \cos\theta}}.$$

From here follows the answer.

Let's introduce the variable $\beta = \pi - \theta$, then $\beta \ll 1$. Method 1 Taking into account the smallness of β, the expression for energy will be written as:

$$E = \frac{A\dot{\beta}^2}{2} + mgl\frac{\beta^2}{2}$$

Такой вид уравнения соответствует гармоническим колебаниям с периодом

$$T = 2\pi\sqrt{\frac{A}{mgl}}.$$

2 метод Введём $\beta_0 = \pi - \theta_0$. Преобразуем интеграл

$$\int\limits_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0 - \cos\theta}}$$

в рассматриваемом приближении.

$$\int\limits_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0 - \cos\theta}} = \int\limits_{\beta_0}^{0}\frac{-d\beta}{\sqrt{\cos(\pi - \beta_0) - \cos(\pi - \beta)}} = \int\limits_{0}^{\beta_0}\frac{d\beta}{\sqrt{\cos\beta - \cos\beta_0}} \simeq\simeq \sqrt{2}\int\limits_{0}^{\beta_0}\frac{d\beta}{\sqrt{\beta_0^2 - \beta^2}} = \sqrt{2}\int\limits_{0}^{\beta_0}\frac{\frac{d\beta}{\beta_0}}{\sqrt{1 - \frac{\beta^2}{\beta_0^2}}} = \sqrt{2}\int\limits_{0}^{1}\frac{dx}{\sqrt{1 - x^2}} = \sqrt{2}\cdot$$

Подстановка найденного значения в исходное выражение для T leads to the answer obtained method 1.

1 method

Taking into account the smallness of β, the expression for energy will be written in the form:

$$E = \frac{A\dot{\beta}^2}{2} + mgl\frac{\beta^2}{2}$$

Такой вид уравнения соответствует гармоническим колебаниям с периодом

$$T = 2\pi\sqrt{\frac{A}{mgl}}.$$

Method 2

Let's enter $\beta_0 = \pi - \theta_0$. Let's transform the integral

$$\int\limits_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0 - \cos\theta}}$$

in the considered approximation.

$$\int_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0-\cos\theta}}=\int_{\beta_0}^{0}\frac{-d\beta}{\sqrt{\cos(\pi-\beta_0)-\cos(\pi-\beta)}}=\int_{0}^{\beta_0}\frac{d\beta}{\sqrt{\cos\beta-\cos\beta_0}}\simeq\simeq\sqrt{2}\int_{0}^{\beta_0}\frac{d\beta}{\sqrt{\beta_0^2-\beta^2}}=\sqrt{2}\int_{0}^{\beta_0}\frac{\frac{d\beta}{\beta_0}}{\sqrt{1-\frac{\beta^2}{\beta_0^2}}}=\sqrt{2}\int_{0}^{1}\frac{dx}{\sqrt{1-x^2}}=\sqrt{2}\cdot$$

Подстановка найденного значения в исходное выражение для T results in the answer obtained in method 1.

Answer:

$$T=4\sqrt{\frac{A}{2mgl}}\int_{\theta_0}^{\pi}\frac{d\theta}{\sqrt{\cos\theta_0-\cos\theta}}$$

D2 **Express L_1 and L_2 in terms of A, C, ω_0.** **0.20**

In a given situation $L_1=L_2$. Their values are instantly obtained from the expression $\vec{L}$ in the principal axes.

Answer:

$$L_1=L_2=C\omega_0$$

D3 **Find a condition relating A, C, m, g, l, ω_0 such that the position $\theta=0$ is stable for the deviations described above. This condition is called Maievsky's condition. It turns out that if the found condition is satisfied, the equilibrium position is stable even if the conservation condition L_1 and L_2 is abandoned.** **0.50**

At $L_1=L_2=C\omega_0$ the expression for the energy becomes

$$E=\frac{A}{2}\dot{\theta}^2+mgl\cos\theta+\frac{C^2\omega_0^2(1-\cos\theta)^2}{2A\sin^2\theta}.$$

Разложим с учётом малости θ:

$$E=\frac{A\dot{\theta}^2}{2}+\left(\frac{C^2\omega_0^2}{4A}-mgl\right)\frac{\theta^2}{2}.$$

Для устойчивости необходимо, чтобы коэффициент перед θ^2 was positive, which determines the required condition.

Answer:

$$\frac{C^2\omega_0^2}{4A}>mgl$$

D4 **Obtain an equation for θ that is satisfied by the angle θ_1. The equation may contain A, C, L_1, L_2, m, g, l. Clue. $U(\theta)$ can be represented as a function of the new variable $s=\cos\theta$, which greatly simplifies the solution.** **0.70**

Clue. $U(\theta)$ can be represented as a function of the new variable $s=\cos\theta$, which greatly simplifies the solution.

In the equilibrium position $\frac{d}{d\theta}U(\theta)=0$. Let's move on to the variable $s=\cos\theta$. Then

$$\frac{d}{d\theta}U(\theta)=\frac{d}{ds}U(s)\cdot\frac{ds}{d\theta}=\frac{d}{ds}U(s)\cdot\sin\theta.$$

Отсюда следует, что условие

$$\frac{d}{d\theta}U(\theta)=0$$

эквивалентно условию

$$\frac{d}{ds}U(s)=0$$

для всех $\theta\in(0,\pi/2)$. Получим $\frac{d}{ds}U(s)$

$$U(s)=mgls+\frac{(L_1-L_2s)^2}{2A\left(1-s^2\right)}\;\frac{d}{d\theta}U(s)=mgl-\frac{(L_1-L_2s)L_2}{A(1-s^2)}+\frac{(L_1-L_2s)^2}{A(1-s^2)^2}.$$

Упростим выражение и перейдём обратно к переменной θ. Equating it to zero, we obtain the required equation.

Answer:

$$\frac{(L_2\cos\theta - L_1)(L_2 - L_1\cos\theta)}{\sin^4\theta} + Amgl = 0$$

D5 **Find the possible precession rates of the top ω_p at $\theta = \theta_1$. Express your answer using A, C, m, g, l, ω_s, θ_1. One of the solutions can be to substitute expressions for L_1, L_2 into the equation from D4 through the angular velocities ω_s, ω_p.** **0.80**

Substituting the expressions for L_1 and L_2 from point B3 into the equation obtained in the previous point, we arrive at a quadratic equation for ω_p:

$$\omega_p^2(A - C)\cos\theta_1 - \omega_p C\omega_s + mgl = 0.$$

It has two solutions, which is the answer.

Answer:

$$\omega_p = \frac{C\omega_s}{2(A-C)\cos\theta_1}\left(1 \pm \sqrt{1 - \frac{4mgl(A-C)\cos\theta_1}{C^2\omega_s^2}}\right)$$

D6 **Obtain approximate expressions for the frequencies in the case of $mgl \ll C\omega_s^2$. Consider that A and C are quantities of the same order.** **0.30**

Let us expand ω_p using the approximate formula $\sqrt{1+x} \simeq 1 + \frac{x}{2}$ for $x \ll 1$. For a larger root:

For a smaller root:

Note that the frequency ω_p^{slow} coincides with the precession frequency of the “fast” top in the simplified model, that is, when the contribution of precession to the angular momentum is neglected.

Answer:

$$\omega_p^{fast} = \frac{C\omega_s}{(A-C)\cos\theta_1}$$

D7 **Determine the angle α_{max} of the maximum deviation of the top axis from the horizontal during the subsequent movement. Express your answer in terms of A, C, m, g, l, ω_0.** **0.70**

Let us write the expression for energy taking into account the initial conditions:

$$L_1 = 0, L_2 = C\omega_0. E = \frac{A\dot{\theta}^2}{2} + mgl\cos\theta + \frac{C^2\omega_0^2\cos^2\theta}{2A\sin^2\theta} = 0.$$

Перейдём к переменной α по формуле:

$$\theta = \frac{\pi}{2} + \alpha.$$

Тогда

$$\frac{A\dot{\alpha}^2}{2} - mgl\sin\alpha + \frac{C^2\omega_0^2\sin^2\alpha}{2A\cos^2\alpha} = 0.$$

Условие $\alpha = \alpha_{max}$ соответствует $\dot{\alpha} = 0$. Подставляя $\dot{\alpha} = 0$ в уравнение выше и преобразовывая, получаем квадратное уравнение на синус максимального угла отклонения:

$$\sin^2\alpha_{max} + \frac{C^2\omega_0^2}{2Amgl}\sin\alpha_{max} - 1 = 0.$$

Решим полученное квадратное уравнение:

$$\sin\alpha_{max} = \pm\sqrt{1 + \left(\frac{C^2\omega_0^2}{4Amgl}\right)^2} - \frac{C^2\omega_0^2}{4Amgl}$$

Так как $|\sin\alpha_{max}| \leq 1$, следует выбрать корень с «+".

Answer:

$$\alpha_{max} = \arcsin\left(\sqrt{1 + \left(\frac{C^2\omega_0^2}{4Amgl}\right)^2} - \frac{C^2\omega_0^2}{4Amgl}\right)$$

D8 **Find the dependence of the angle of deviation of the top axis from the horizontal on time $\alpha(t)$. Express your answer in terms of A, C, m, g, l, ω_0.** **0.70**

Let us write an approximate expression for α_{max} in this case:

$$\alpha_{max} = \frac{2Amgl}{C^2\omega_0^2} \sim \frac{mgl}{C\omega_0^2} \ll 1.$$

В силу малости α_{max} можем рассматривать выражение для энергии для малых α. С учётом начальных условий полная энергия равна нулю.

$$\frac{A\dot{\alpha}^2}{2} + \frac{C^2\omega_0^2}{2A}\alpha^2 - mgl\alpha = 0$$

Такой вид выражения для энергии соответствует колебаниям с частотой

$$\omega = \frac{C}{A}\omega_0,$$

амплитудой

$$\alpha_0 = \frac{Amgl}{C^2\omega_0^2}$$

и смещением от центра

$$\alpha_1 = \alpha_0 = \frac{Amgl}{C^2\omega_0^2}.$$

С учётом всех найденных параметров движения и начальных условий получаем зависимость $\alpha(t)$.

Answer:

$$\alpha(t) = \frac{Amgl}{C^2\omega_0^2}\left(1 - \cos\left(\frac{C}{A}\omega_0 t\right)\right)$$

D9 **Find the dependence of the precession angle on time $\varphi(t)$. At the initial moment $\varphi(0) = \varphi_0$. Express your answer using A, C, m, g, l, ω_0, φ_0.** **0.50**

Let's express $\dot{\varphi}(t)$ using the found dependence $\alpha(t)$:

$$\dot{\varphi} = \frac{L_1 - L_2\cos\theta}{A\sin^2\theta} = \frac{C\omega_0\sin\alpha}{A\cos^2\alpha} \simeq \frac{mgl}{C\omega_0}\left(1 - \cos\left(\frac{C}{A}\omega_0 t\right)\right)$$

Integrating, we get the answer.

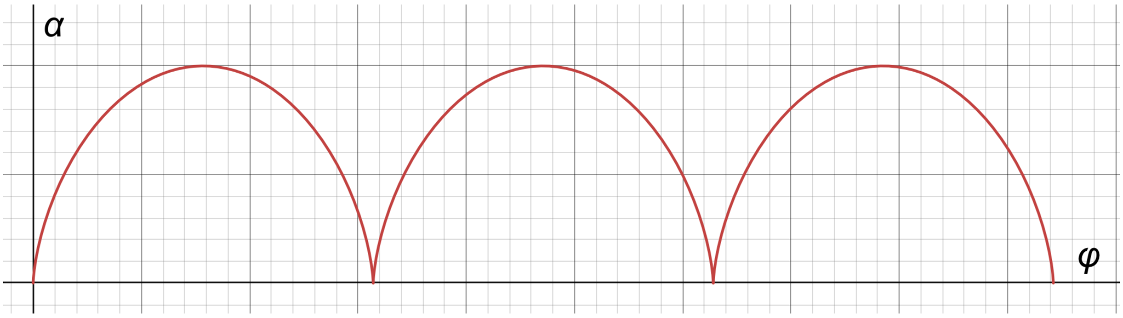
Answer:

$$\varphi(t) = \varphi_0 + \frac{mgl}{C\omega_0}\left(t - \frac{A}{C\omega_0}\sin\left(\frac{C}{A}\omega_0 t\right)\right)$$

D10 **Draw a characteristic graph of the dependence $\alpha(\varphi)$. Determine the angle $\Delta\varphi$ through which the top axis rotates between two successive returns of the top axis to a horizontal position. Express your answer using A, C, m, g, l, ω_0.** **0.50**

Expressions $\varphi(t)$, $\alpha(t)$ parametrically define a cycloid, the characteristic form of which is presented in the figure below.

The time between subsequent instants $\alpha = 0$ is $t = \dfrac{2\pi}{\omega_0}\dfrac{A}{C}$. Substituting t into the expression for φ gives the answer.



D11 **Calculate $\Delta\varphi$ for the described system.** **0.50**

Let's find A, C, l:

$$C = \frac{(2m)R^2}{2} = mR^2 A = \frac{(2m)R^2}{4} + (2m)R^2 + \frac{mR^2}{3} = \frac{17}{6} mR^2 l = \frac{2mR + m\frac{R}{2}}{3m} = \frac{5}{6} R$$

Подставим в формулу для $\Delta\varphi$ найденные величины (требуется также учесть, что масса волчка составляет $3m$) and get the answer.

Answer:

$$\Delta\varphi = \frac{17}{120}\pi$$